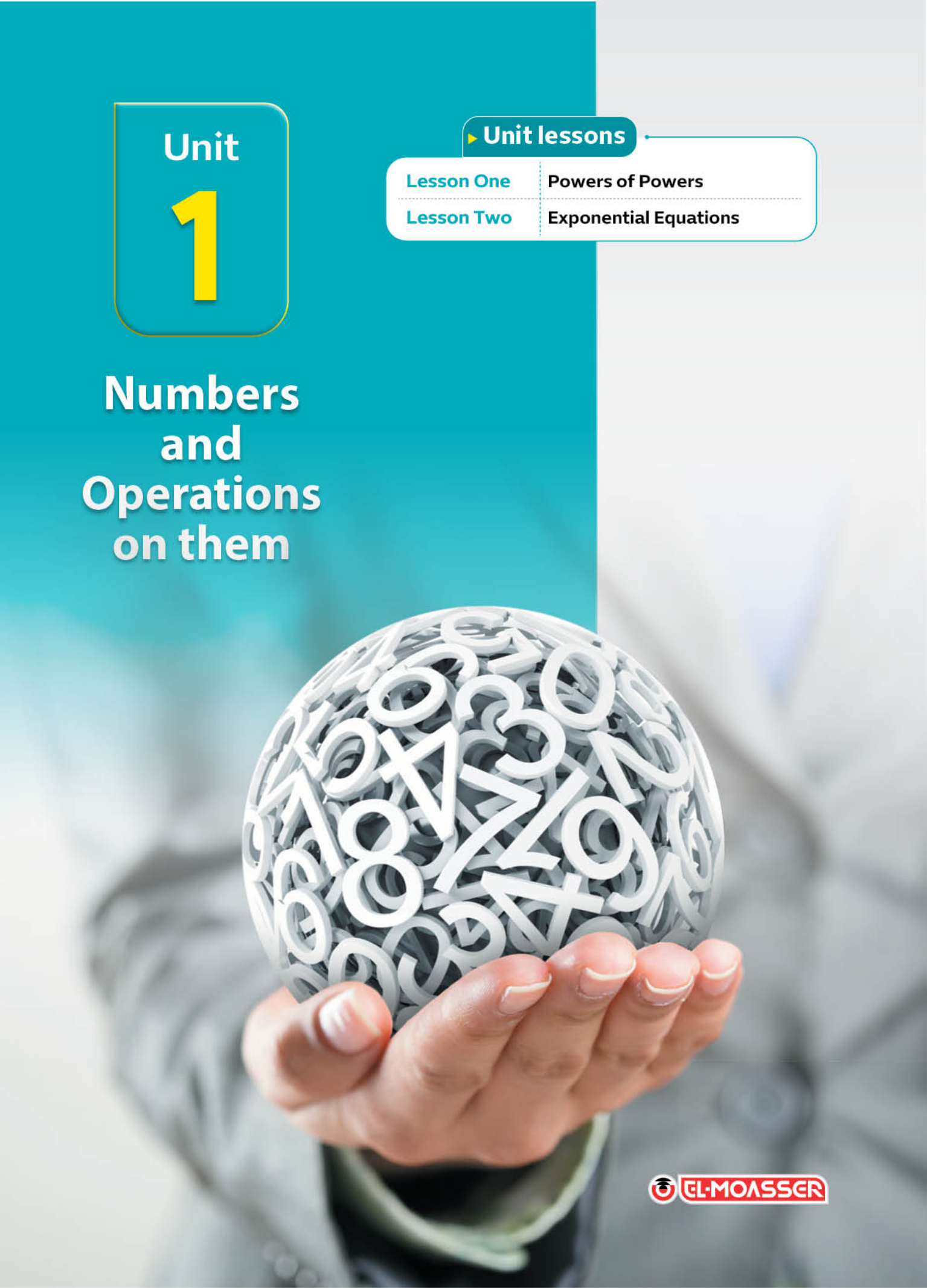
Unit

1

Numbers and Operations on them

Unit lessons

Lesson One	Powers of Powers
Lesson Two	Exponential Equations



EL-MOASSER

Lesson ONE

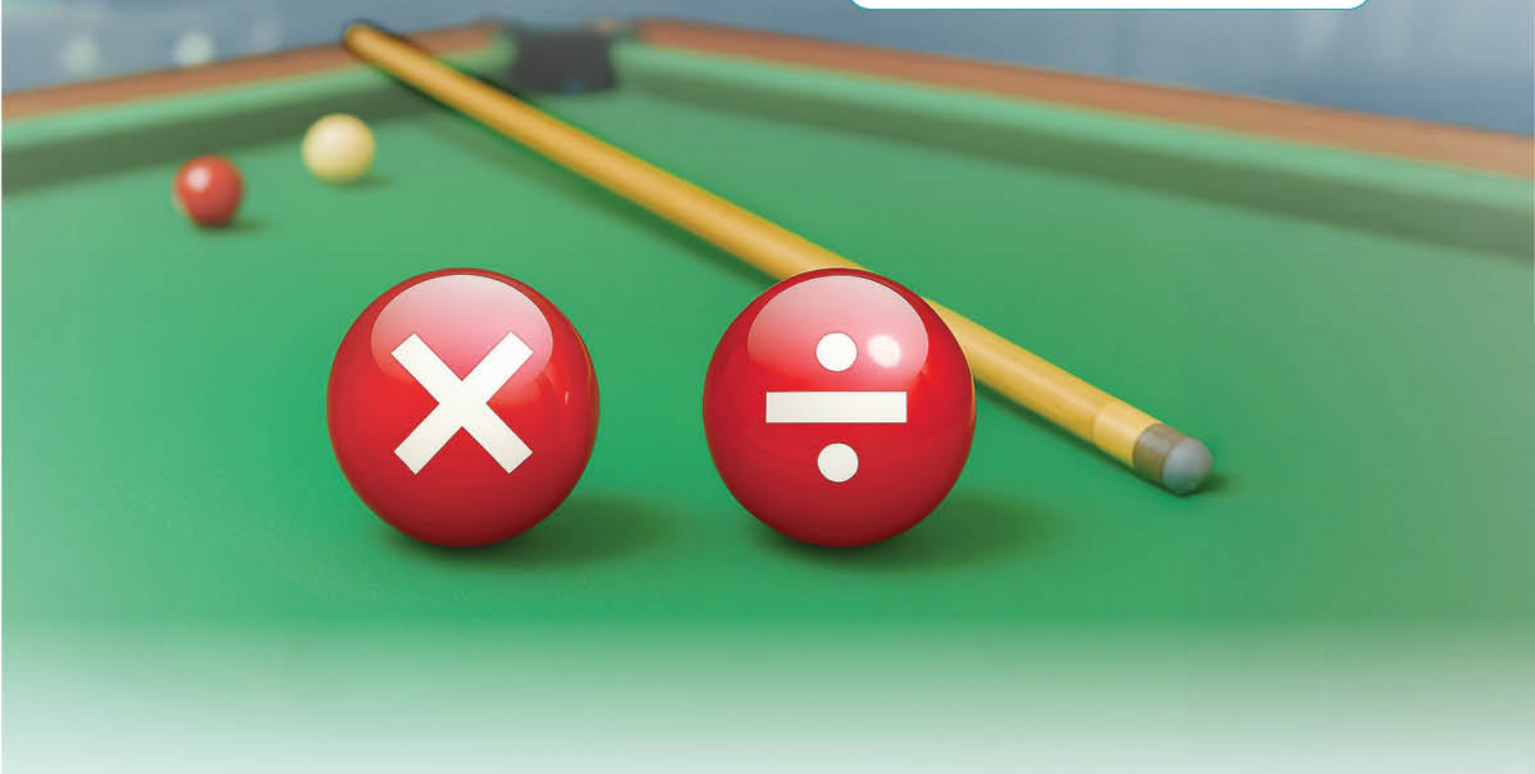
Powers of Powers

Learning Outcomes

- Learn the concept of a power of the power.
- Find the power of a power for a real number.
- Use the power of a power rule to simplify mathematical expressions.
- Use the power of a product and a quotient rule to solve real-life problems.

Vocabulary

- Exponent
- Base
- Power
- Power of a power



Lesson Map

Remember the laws of exponents in real numbers → Power of a power rule → Power of a product of two powers rule and power of a quotient of two powers rule

EL-MOASSER

Lesson One

Powers of Powers



Remember the laws of exponents in real numbers

If a , b are real numbers and m , n are two integers , taking into account the exceptions where the denominator is equal to zero , and the cases where both the base and the exponent are equal to zero , then :

Rule	Example	Explanation
$a^m \times a^n = a^{m+n}$	• $4^3 \times 4^2 = 4^{3+2} = 4^5$	When multiplying powers with the same base, keep the base and add the exponents
$\frac{a^m}{a^n} = a^{m-n}$	• $\frac{3^6}{3^2} = 3^{6-2} = 3^4$	When dividing powers with the same base, keep the base and subtract the exponents
$(ab)^n = a^n b^n$	• $(3 \times 4)^2 = 3^2 \times 4^2$	When finding the power of a product of two numbers, distribute the exponent over the two numbers.
$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$	• $\left(\frac{2}{5}\right)^3 = \frac{2^3}{5^3}$	When finding the power of a quotient of two numbers, distribute the exponent over the two numbers
$a^0 = 1$	• $\left(\sqrt{3}\right)^0 = 1$, $(-5)^0 = 1$	A non-zero real number raised to the power of zero equals 1
$a^{-n} = \frac{1}{a^n}$	• $3^{-2} = \frac{1}{3^2} = \frac{1}{9}$	A non-zero real number raised to the power of (– n) where n is an integer equals the multiplicative inverse of the number itself raised to the power of n

Power of a power rule

$$(5^2)^3 = \overbrace{5^2 \times 5^2 \times 5^2}^{\text{3 factors}} = 5^{2+2+2} = 5^6 \leftarrow (2 \times 3)$$

Note that: 6 is the product of the two original exponents: 2 , 3 ($2 \times 3 = 6$)

That means : To find the power of a power , multiply the exponents.

In general

For any non-zero real number a and any two integers m and n , then :

$$(a^m)^n = a^{m \times n}$$

For example : • $(2^4)^2 = 2^{4 \times 2} = 2^8$ • $(3^{-2})^5 = 3^{-2 \times 5} = 3^{-10}$

Example 1

Simplify each of the following to its simplest form :

1 $(2^3)^5 \times (2^{-2})^6$

2 $\dfrac{(3^5)^6 \times (3^{-3})^4}{3^7 \times (3^2)^5}$

3 $\dfrac{\left((-\sqrt{2})^2\right)^3 \times (\sqrt{2})^9}{\left((-\sqrt{2})^5\right)^3}$

Solution

1 $(2^3)^5 \times (2^{-2})^6 = 2^{3\times5} \times 2^{-2\times6} = 2^{15} \times 2^{-12} = 2^{15-12} = 2^3 = 8$

2 $\dfrac{(3^5)^6 \times (3^{-3})^4}{3^7 \times (3^2)^5} = \dfrac{3^{5\times6} \times 3^{-3\times4}}{3^7 \times 3^{2\times5}} = \dfrac{3^{30} \times 3^{-12}}{3^7 \times 3^{10}} = \dfrac{3^{30-12}}{3^{7+10}} = \dfrac{3^{18}}{3^{17}} = 3^{18-17} = 3$

3 $\dfrac{\left((-\sqrt{2})^2\right)^3 \times (\sqrt{2})^9}{\left((-\sqrt{2})^5\right)^3} = \dfrac{\left((\sqrt{2})^2\right)^3 \times (\sqrt{2})^9}{\left(-(\sqrt{2})^5\right)^3}$

$= \dfrac{(\sqrt{2})^6 \times (\sqrt{2})^9}{-(\sqrt{2})^{15}}$

$= \dfrac{(\sqrt{2})^{15}}{-(\sqrt{2})^{15}} = -1$

Remember that

- $(-a)^n = a^n$, if n is an even integer
- $(-a)^n = -a^n$, if n is an odd integer

TRY it Yourself 1

Simplify each of the following to its simplest form :

1 $\dfrac{(5^4)^{-2} \times (5^3)^3}{(5^2)^2 \times (5^{-2})^3}$

2 $\dfrac{\left((-\sqrt{3})^3\right)^5 \times \left((\sqrt{3})^{-4}\right)^3}{\left((\sqrt{3})^2\right)^3}$

Example 2

Simplify to the simplest form : $\dfrac{(x^3)^2 \times x^{-4}}{(x^6)^3 \times (x^{-9})^2}$ **, then find the value of the result when** $x = 2$

Solution

$\dfrac{(x^3)^2 \times x^{-4}}{(x^6)^3 \times (x^{-9})^2} = \dfrac{x^6 \times x^{-4}}{x^{18} \times x^{-18}} = \dfrac{x^{6-4}}{x^{18-18}} = x^2$

When $x = 2$: $\therefore$ The expression $= 4$

Note that

Division by zero is meaningless, and therefore, when there are symbols in the denominator, these symbols are not equal to zero.

TRY it Yourself 2

Simplify to the simplest form : $\dfrac{(n^{-2})^5 \times (n^3)^5}{(n^{-1})^4 \times (n^2)^3}$ **, then find the value of the result when** $n = -1$

Example 3

Simplify to the simplest form : $\frac{(81)^x \times (6^2)^x}{(27)^{2x-1} \times 4^x}$

Solution

$$\frac{(81)^x \times (6^2)^x}{(27)^{2x-1} \times 4^x} = \frac{(3^4)^x \times 6^{2x}}{(3^3)^{2x-1} \times (2^2)^x} = \frac{3^{4x} \times (2 \times 3)^{2x}}{3^{6x-3} \times 2^{2x}} = \frac{3^{4x} \times 2^{2x} \times 3^{2x}}{3^{6x-3} \times 2^{2x}}$$

$$= 3^{4x+2x-6x+3} \times 2^{2x-2x} = 3^3 \times 2^0 = 27$$

TRY it Yourself 3

Simplify to the simplest form : $\frac{8^{2x-1} \times 9^{3x-2}}{6^{6x-3}}$

Example 4

Choose the correct answer from the given ones :

1 $(3^2)^5 =$ .....

(a) 3^{2^5} (b) 3^{25} (c) 3^{10} (d) 3^7

2 $(2+2)^3 = 2^{.....}$

(a) 4 (b) 5 (c) 6 (d) 8

3 If $5^a = 4$, what is the value of : $25^a + 125^a$?

(a) 4^a (b) $(25 + 125^a)$ (c) 150 (d) 80

4 If $6 = 2^{x+1}$, what is the value of : 4^x ?

(a) 16 (b) 9 (c) 36 (d) 25

Solution

1 **(c) The reason :** $(3^2)^5 = 3^{2 \times 5} = 3^{10}$

2 **(c) The reason :** $(2+2)^3 = (4)^3 = (2^2)^3 = 2^6$

3 **(d) The reason :** $25^a + 125^a = (5^2)^a + (5^3)^a$

$= (5^a)^2 + (5^a)^3 = 4^2 + 4^3$

$= 16 + 64 = 80$

Note that

① $(a^m)^n = (a^n)^m = a^{mn}$

② $(a^m)^n \neq a^{m^n}$

4 **(b) The reason :** $\because 6 = 2^{x+1}$ $\therefore 6 = 2^x \times 2$

$\therefore 2^x = 6 \div 2$ $\therefore 2^x = 3$

$\therefore 4^x = (2^2)^x = (2^x)^2 = 3^2 = 9$

TRY it Yourself 4

Choose the correct answer from the given ones :

1 $(3 + 3 + 3)^4 = 3^{\dots\dots}$

(a) 6 (b) 12 (c) 8 (d) 5

2 If $7^x = 3$, then $49^x = \dots\dots\dots\dots$

(a) 21 (b) 7^2 (c) 9 (d) 98

3 If $50 = 5^{x+2}$, then $5^x = \dots\dots\dots\dots$

(a) 10 (b) 5 (c) 2 (d) 3

Power of a product of two powers rule and power of a quotient of two powers rule

Power of a product of two powers rule

For any two non-zero real numbers a and b and three integers m , n and k , then :

$$(a^m \times b^n)^k = a^{mk} \times b^{nk}$$

For example :

$$(3^4 \times 5^2)^3 = 3^{4\times3} \times 5^{2\times3}$$
$$= 3^{12} \times 5^6$$

Power of a quotient of two powers rule

$$\left(\frac{a^m}{b^n}\right)^k = \frac{a^{mk}}{b^{nk}}$$

For example :

$$\left(\frac{3^3}{5^2}\right)^5 = \frac{3^{3\times5}}{5^{2\times5}} = \frac{3^{15}}{5^{10}}$$

Example 5

Simplify each of the following to its simplest form :

1 $(5^{-2}\, a^{-3}\, b^4)^3 \times (5\, a^3\, b^{-2})^4$

2 $\left(\frac{x^2 y^5}{x^4 y^2}\right)^3$

Solution

1 $(5^{-2}\, a^{-3}\, b^4)^3 \times (5\, a^3\, b^{-2})^4 = 5^{-2\times3} \times a^{-3\times3} \times b^{4\times3} \times 5^4 \times a^{3\times4} \times b^{-2\times4}$

$= 5^{-6}\, a^{-9}\, b^{12} \times 5^4\, a^{12}\, b^{-8}$

$= 5^{-6+4}\, a^{-9+12}\, b^{12-8} = 5^{-2}\, a^3\, b^4 = \frac{a^3 b^4}{25}$

---

2 $\left(\frac{x^2 y^5}{x^4 y^2}\right)^3 = \left(\frac{y^3}{x^2}\right)^3 = \frac{y^{3\times3}}{x^{2\times3}} = \frac{y^9}{x^6}$

TRY it Yourself 5

Simplify each of the following to its simplest form :

1 $(3\, x^{-2}\, y^{-3})^5\, (x^2\, y^4)^4$

2 $\left(\frac{2\, m^{-3}\, n^4}{n^5\, m^{-4}}\right)^{-3}$

Example 6

Simplify to the simplest form : $\dfrac{(36)^{x-2} \times (48)^{x+1}}{(18)^{6x-4}}$

, then find the value of the result when : $x = 1$

Solution

$$\frac{(36)^{x-2} \times (48)^{x+1}}{(18)^{6x-4}} = \frac{(2^2 \times 3^2)^{x-2} \times (3 \times 2^4)^{x+1}}{(2 \times 3^2)^{6x-4}}$$

$$= \frac{2^{2x-4} \times 3^{2x-4} \times 3^{x+1} \times 2^{4x+4}}{2^{6x-4} \times 3^{12x-8}} = \frac{2^{6x} \times 3^{3x-3}}{2^{6x-4} \times 3^{12x-8}}$$

$$= 2^{6x-6x+4} \times 3^{3x-3-12x+8} = 2^4 \times 3^{5-9x}$$

• When $x = 1$

$$= 2^4 \times 3^{5-9(1)} = 2^4 \times 3^{-4} = \left(\frac{2}{3}\right)^4 = \frac{16}{81}$$

TRY it Yourself 6

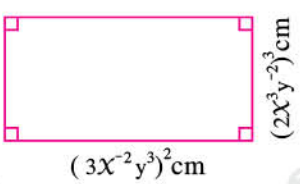
Simplify to the simplest form : $\dfrac{(3^n)^3 \times (45)^{n+2}}{(3^{2n})^3 \times 5^{n+2}}$

, then find the value of the result when : $n = 1$

Example 7

Write the area of the given rectangle

in its simplest form.



Solution

Area of the rectangle = Length × Width

$$= (2x^3y^{-2})^3 \times (3x^{-2}y^3)^2$$

$$= 2^3 x^9 y^{-6} \times 3^2 x^{-4} \times y^6$$

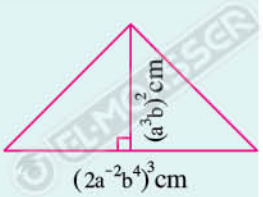
$$= 8 \times 9 x^{9-4} y^{-6+6}$$

$$= 72x^5$$

TRY it Yourself 7

Write the area of the given triangle

in its simplest form.



Exercise 1 — Powers of Powers



Remember · Understand · Apply · Problems Solving | From the school book

Interactive test

1 Find the value of each of the following in its simplest form :

1 $(5^3)^{-2} \times (5^2)^2$

2 $\left((-7)^3\right)^5 (7^2)^{-7}$

3 $\dfrac{(2^{11})^2 \times 2^3}{(2^6)^3 \times 2^4}$

4 $\dfrac{(-3^5)^6 \times 3^2}{3^{-2} \times (3^6)^6}$

5 $\dfrac{4^{x+2} \times 9^{3+x}}{6^{2x+3}}$

6 $\dfrac{\left((\sqrt{5})^3\right)^2 \times \sqrt{5}}{(-\sqrt{5})^5}$

7 $\left(\sqrt{5}\right)^3 \times \left(\left(-\sqrt{5}\right)^2\right)^3$

8 $(-5^2)^3 \times \left(\left(-\sqrt{5}\right)^{-4}\right)^2$

9 $\dfrac{\left((\sqrt{7})^{-1}\right)^3 \times \left((-\sqrt{7})^4\right)^3}{\left((\sqrt{7})^2\right)^3}$

2 Simplify each of the following to its simplest form :

1 $(4\,a^4\,b^3\,c)^2$

2 $(-2\,m^5\,n^4)^3$

3 $(c^{-2})^3 \times (-2\,c^5)^2$

4 $(a^3)^5 \times 5^2$

5 $(m^2)^{-3} \times m^6$

6 $(3\,x\,y^4)^{-2} \times \left((-2\,y)^2\right)^3$

7 $(-3\,x^2\,y^3\,z)^2 \times \left((-x^2\,y)^3\right)^2$

8 $\dfrac{(a^4)^3}{(a^2)^5}$

9 $\left(\dfrac{2\,a^2\,b^2}{b^{-1}\,a^4}\right)^3$

10 $\dfrac{(3^{-2}\,m^3\,n^2)^{-2}}{(6\,m^{-2}\,n^{-1})^3}$

11 $\left(\dfrac{2\,p^2\,s^{-2}}{4^{-2}\,p^4\,s^3}\right)^{-2}$

12 $\dfrac{(16\,x^2\,y^{-2})^0}{(4\,x^0\,y^{-6})^{-2}}$

13 $\left(\dfrac{3^0\,c^3\,d^4\,f^2}{2\,c^4\,d^3}\right)^{-5}$

14 $\dfrac{(2\,a^5)^3 \times (3\,a^3)^2}{(6\,a^7)^2 \times (a^{-1})^3}$

15 $\dfrac{(x^2)^{-3} \times (x^{-1})^2}{x^{-3} \times x^{-4}}$

16 $\dfrac{y^8 \times (y^{-3})^2}{(y^5)^2 \times (y^{-2})^6}$

3 Simplify each of the following to its simplest form :

1 $\left(\left(\left(\sqrt{3}\right)^3\right)^{-2}\right)^{-1}$ « 27 »

2 $\dfrac{36^n \times 5^n}{30^{2n}}$ « $\frac{1}{5^n}$ »

3 $\dfrac{(12)^5 \times 9^3}{8^2 \times 6^4}$ « 2187 »

4 $\dfrac{\left((\sqrt{6})^7\right)^3 \times \left((\sqrt{2})^{-5}\right)^5}{\left((\sqrt{3})^4\right)^5 \times \left((\sqrt{2})^2\right)^{-2}}$ « $\sqrt{3}$ »

4 **Simplify each of the following to its simplest form , then find the value of the result when $x = 1$:**

1 📖 $\dfrac{4^{x} \times 2^{x-4}}{16^{x-2}}$ « 8 »

2 $\dfrac{4^{x} \times 9^{2-x}}{6^{2x}}$ « 1 »

3 📖 $\dfrac{(4)^{x} \times (49)^{x-1}}{(98)^{x}}$ « $\frac{2}{49}$ »

4 📖 $\dfrac{\left((\sqrt{5})^{3}\right)^{2x} \times (15)^{x-1}}{(75)^{x}}$ « $\frac{5}{3}$ »

5 📖 **Simplify to the simplest form :** $\dfrac{9^{2n-1} \times 18^{1-4n}}{36^{2-3n}}$ **, then find the numerical value of the result when** $n = 2$ « 2 »

6 **Prove that :**

1 📖 $\dfrac{27^{x-1} \times 8^{x}}{(2\sqrt{2})^{2x} \times (3\sqrt{3})^{2x}} = \dfrac{1}{27}$

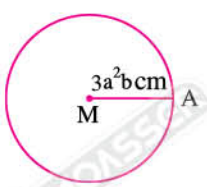
2 $\dfrac{(3^{2}x^{4})^{-3} \times (6^{3}x^{2}y^{-1})^{4}}{6^{6}(x^{-2}y^{-2})^{2}} = 64$

7 📖 **Geometry : Find the area of each figure in its simplest form :**

1

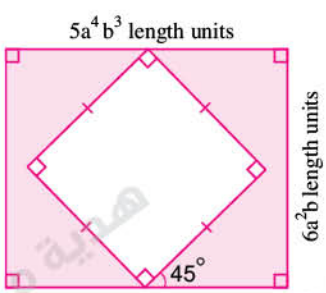


2

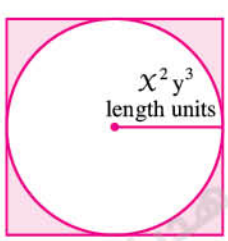


8 **Write the area of the shaded part in each of the following in its simplest form :**

1



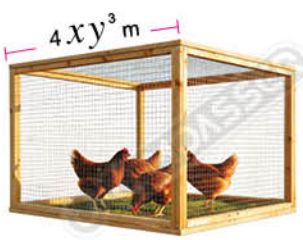
2



9 📖 **Decoration :** A room has a square floor of side length $2x^3y^2$ length units. The floor will be tiled using square tiles. The length of one side of each tile is xy^2 length units. How many tiles are needed to cover the whole floor ? « $4x^4$ tiles »

10 📖 **Animal wealth :** A chicken coop is in the shape of a cube. The length of its edge is $4xy^3$ meters. Find the volume of the coop in its simplest form.

« $64x^3y^9$ m^3 »



11 📖 **Energy Sources :** A cylindrical barrel is used to store fuel. The radius of its base is $2\ kw^2$ meters and its height is 4 k meters.
Find the volume of the barrel in its simplest form.



« $16\ \pi\ k^3\ w^4\ m^3$ »

12 If $7^x = 14$, **find the value of :** 49^{x-1} « 4 »

13 📖 If $3^a = 2$, **find the value of :** $9^a + (27)^a$ « 12 »

14 If $3^x = 4$, **find the value of :** $27^x - 5 \times 3^{x+1}$ « 4 »

15 📖 **Spot the mistake :** Khaled and Asmaa both simplified $(a^2)^3 \times a^4$ to its simplest form. Their solutions are shown. Which one is correct ? Discuss.

Khaled's solution

$(a^2)^3 \times a^4 = a^5 \times a^4 = a^{20}$

Asmaa's solution

$(a^2)^3 \times a^4 = a^6 \times a^4 = a^{10}$

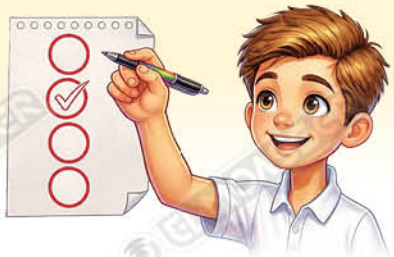
Creative Thinking

16 📖 If $10^x = 5^{x+1}$, **find the value of :** 4^x « 25 »

17 📖 If $a = 5^{30}$, $b = 3^{45}$, $c = 2^{60}$, arrange a , b and c in an ascending order.



Multiple Choice Questions



Choose the correct answer from the given ones :

1 Which of the following equals $(a^4)^7$?

(a) a^3 (b) a^{11}

(c) a^{28} (d) a^{47}

2 What is the value of $(3^8)^0$?

(a) 3^8 (b) 3^{80}

(c) 1 (d) 24

3 Which of the following equals $(x^3)^5$?

(a) x^8 (b) x^2

(c) $(x^5)^3$ (d) $(x^3) \times 5$

4 Which of the following equals $(2^3)^2$?

(a) 2^5 (b) 2^6

(c) 2^{3^2} (d) 2^{2^3}

5 Which of the following equals $(3^4)^2 \times 3^5$?

(a) 3^{11} (b) 3^{13}

(c) 3^{40} (d) 9^{11}

6 Which of the following does not equal 2^{12} ?

(a) 8^4 (b) 4^6

(c) $(16)^3$ (d) 6^5

7 The area of the square whose side length is $4\,a^3\,x^5$ cm in its simplest form is ..... cm^2

(a) $8\,a^6\,x^{10}$ (b) $16\,a^3\,x^5$

(c) $16\,a^6\,x^{10}$ (d) $16\,a^9\,x^{25}$

8 What is the result of $x^2 \times (x^5)^2$ in its simplest form ?

(a) x^9 (b) x^{12}

(c) x^{14} (d) x^{20}

9 What is the result of $\frac{(x^3)^5}{(x^8)^2}$ in its simplest form ?

(a) $\frac{1}{x}$ (b) x

(c) x^{-2} (d) x^{31}

10 What is the result of $(-a^3\,b)^2$ in its simplest form ?

(a) $a^3\,b^3$ (b) $-a^5\,b^3$

(c) $a^6\,b^2$ (d) $-a^6\,b^2$

11 What is the result of $\left(\frac{3\,a^2}{2\,b}\right)^2$ in its simplest form ?

(a) $\frac{9\,a^4}{4\,b^3}$ (b) $\frac{9\,a^4}{4\,b^2}$

(c) $\frac{3\,a^4}{2\,b^2}$ (d) $\frac{3\,a^4}{2\,b^3}$

12 Which of the following equals $\left(\left(\sqrt{3}\right)^5\right)^3$?

(a) $\left(\sqrt{3}\right)^8$ (b) $\left(9\sqrt{3}\right)^3$

(c) 3^6 (d) $\left(\sqrt{3}\right)^{10}$

13 $\left(\frac{b^{-2} c^{0} a^{4}}{3 a^{6} b^{-4}}\right)^{-2} =$ .....

(a) $\left(\frac{b}{3a}\right)^{4}$ (b) $\frac{b^{3}}{4 a^{4}}$

(c) $9\left(\frac{a}{b}\right)^{4}$ (d) $\frac{b^{2}}{3 a^{2}}$

14 If $2^{x} = 5$, what is the value of 8^{x} ?

(a) 125 (b) 32

(c) 27 (d) 15

15 $(4 + 4)^{2} = 2^{.....}$

(a) 3 (b) 5

(c) 6 (d) 8

16 If $\left((10)^{2} \times 8\right)^{4} = 2^{a} \times 5^{b}$, then

$a + b =$ .....

(a) 6 (b) 20

(c) 8 (d) 28

17 If $3 = 5^{x-1}$, then $5^{x} =$ .....

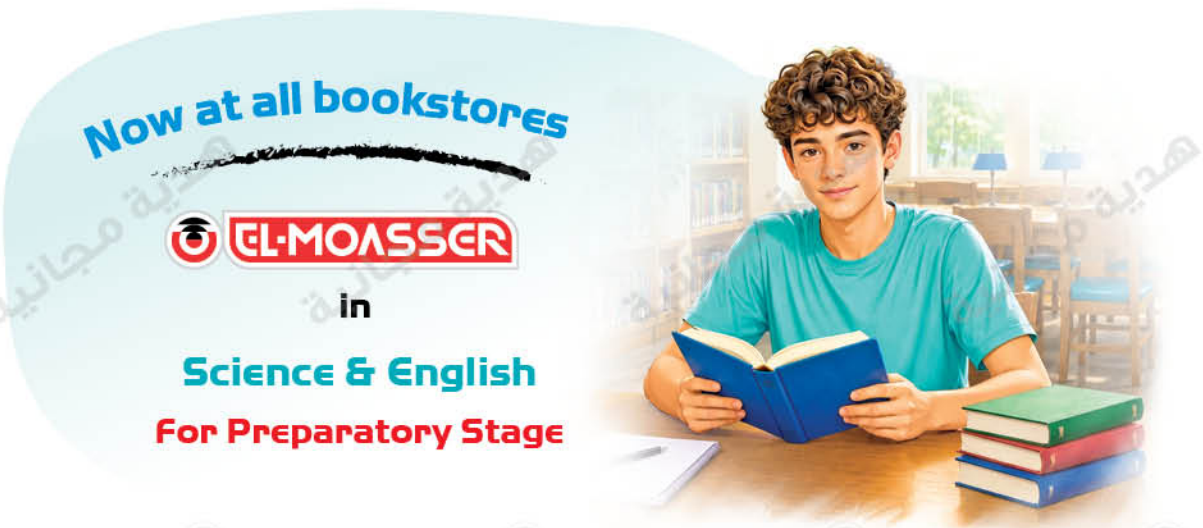
(a) 15 (b) 9

(c) 8 (d) 18

18 $\left(-\sqrt{3}\right)^{3} \times 3 = -\left(\sqrt{3}\right)^{.....}$

(a) 4 (b) 5

(c) 6 (d) 2



Lesson
Two

Exponential Equations

Learning Outcomes

- Solve exponential equations using the laws of exponents.
- Use exponential equations to solve real-life problems.

Vocabulary

- Exponential equation



Lesson Map

Exponential equations → Solving exponential equations

EL-MOASSER

Lesson **Two**

Exponential Equations



Exponential equations

Definition of an exponential equation

It is an equation that contains a variable in the exponent, and our study of the exponential equation will be limited to the form $a^{x} = b$ where $a > 0, a \neq 1$

Examples of exponential equations

- $5^{x} = 25$
- $(4)^{x-3} = 16$
- $7^{2x-1} = \frac{1}{49}$

Examples of non-exponential equations

- $(-3)^{2} = x$
- $x + 3 = 5^{2}$
- $x^{2} - 2 = 23$

Solving Exponential Equations

To solve exponential equations we use the property of equality of powers

If $a^{x} = a^{y}$ where $a > 0$, $a \neq 1$, then $x = y$

Exponential form 2^{3} ⟷ Numerical form 8

For example :

To solve the exponential equation $3^{x} = 81$ → (Write both sides of the equation in exponential form) $3^{x} = 3^{4}$ → (Use the property of equality of powers) $x = 4$

Use factorization

81	3
27	3
9	3
3	3
1	

3^{4}

$81 = 3^{4}$

Example 1

Solve each of the following equations in $\mathbb{R}$:

1 $2^{x+5} = 8$ 2 $3^{2x-1} = \frac{1}{27}$ 3 $\left(\frac{5}{7}\right)^{2x} = \frac{49}{25}$

Solution

1 To solve the equation , write 8 in an exponential form with base 2

$\because 2^{x+5} = 8$

$\therefore 2^{x+5} = 2^{3}$

$\because$ The bases are equal.

$\therefore$ The exponents are equal.

$\therefore x + 5 = 3$

$\therefore x = -2$

8	2
4	2
2	2
1	

2^{3}

2 To solve the equation , write $\frac{1}{27}$ in an exponential form with base 3

$\because 3^{2x-1} = \frac{1}{27}$

$\therefore 3^{2x-1} = \frac{1}{3^{3}}$

$\therefore 3^{2x-1} = 3^{-3}$

$\because$ The bases are equal.

$\therefore$ The exponents are equal.

$\therefore 2x - 1 = -3$ $\therefore 2x = -2$

$\therefore x = -1$

27	3
9	3
3	3
1	

3^{3}

3 $\because \left(\frac{5}{7}\right)^{2x} = \frac{49}{25}$ $\therefore \left(\frac{5}{7}\right)^{2x} = \frac{7^2}{5^2}$

$\therefore \left(\frac{5}{7}\right)^{2x} = \left(\frac{7}{5}\right)^{2}$ $\therefore \left(\frac{5}{7}\right)^{2x} = \left(\frac{5}{7}\right)^{-2}$

$\because$ The bases are equal. $\therefore$ The exponents are equal.

$\therefore 2x = -2$ $\therefore$ $x = -1$

Note that

If a , b are two non-zero real numbers , n is an integer , then : $\left(\frac{a}{b}\right)^{n} = \left(\frac{b}{a}\right)^{-n}$

For example : $\left(\frac{7}{5}\right)^{2} = \left(\frac{5}{7}\right)^{-2}$

TRY it Yourself 1

Solve each of the following equations in $\mathbb{R}$:

1 $5^{n} = 125$ 2 $2^{2n-3} = \frac{1}{8}$ 3 $\left(\frac{3}{2}\right)^{x-2} = \frac{16}{81}$

Example 2

Solve each of the following equations in $\mathbb{R}$:

1 $(16)^{x-2} = 2^{2x}$ 2 $(81)^{x-1} = (27)^{x+2}$ 3 $2^{x^2-1} = 8^{x+1}$

Solution

1 $\because (16)^{x-2} = 2^{2x}$ $\therefore (2^4)^{x-2} = 2^{2x}$

$\therefore 2^{4x-8} = 2^{2x}$

$\because$ The bases are equal. $\therefore$ The exponents are equal.

$\therefore 4x - 8 = 2x$ $\therefore 2x = 8$ $\therefore$ $x = 4$

16	2
8	2
4	2
2	2
1	

2^4

2 $\because (81)^{x-1} = (27)^{x+2}$ $\therefore (3^4)^{x-1} = (3^3)^{x+2}$

$\therefore 3^{4x-4} = 3^{3x+6}$

$\because$ The bases are equal. $\therefore$ The exponents are equal.

$\therefore 4x - 4 = 3x + 6$ $\therefore$ $x = 10$

81	3
27	3
9	3
3	3
1	

3^4

27	3
9	3
3	3
1	

3^3

3 $\because 2^{x^2-1} = 8^{x+1}$ $\therefore 2^{x^2-1} = (2^3)^{x+1}$ $\therefore 2^{x^2-1} = 2^{3x+3}$

$\therefore x^2 - 1 = 3x + 3$ $\therefore x^2 - 3x - 4 = 0$ $\therefore (x-4)(x+1) = 0$

$\therefore x - 4 = 0$ or $\therefore x + 1 = 0$

$\therefore$ $x = 4$ | $\therefore$ $x = -1$

TRY it Yourself 2

Solve each of the following equations in $\mathbb{R}$:

1 $(125)^{3x-2} = 5^{12}$ 2 $32^{2x-1} = 8^{3x+2}$ 3 $3^{x^2} = 81$

Example 3

Solve each of the following equations in $\mathbb{R}$:

1 $\left(\frac{8}{125}\right)^{-1} = \left(\frac{5}{2}\right)^{|x|}$ 2 $\left(\frac{5}{3}\right)^{3-5x} = \left(\frac{81}{625}\right)^{x+1}$

Solution

1 $\because \left(\frac{8}{125}\right)^{-1} = \left(\frac{5}{2}\right)^{|x|}$ $\therefore \left(\frac{2^3}{5^3}\right)^{-1} = \left(\frac{5}{2}\right)^{|x|}$ $\therefore \frac{5^3}{2^3} = \left(\frac{5}{2}\right)^{|x|}$

$\therefore \left(\frac{5}{2}\right)^3 = \left(\frac{5}{2}\right)^{|x|}$ $\therefore |x| = 3$ $\therefore x = \pm 3$

---

2 $\because \left(\frac{5}{3}\right)^{3-5x} = \left(\frac{81}{625}\right)^{x+1}$ $\therefore \left(\frac{5}{3}\right)^{3-5x} = \left(\frac{3^4}{5^4}\right)^{x+1}$

$\therefore \left(\left(\frac{3}{5}\right)^{-1}\right)^{3-5x} = \left(\left(\frac{3}{5}\right)^4\right)^{x+1}$ $\therefore \left(\frac{3}{5}\right)^{-3+5x} = \left(\frac{3}{5}\right)^{4x+4}$

$\therefore -3+5x = 4x+4$ $\therefore 5x-4x = 4+3$ $\therefore x = 7$

TRY it Yourself 3

Solve each of the following equations in $\mathbb{R}$:

1 $\left(\frac{49}{9}\right)^{-2} = \left(\frac{3}{7}\right)^{|x|}$ 2 $\left(\frac{5}{7}\right)^{8-2x} = \left(\frac{25}{49}\right)^{2x+1}$

Notes

- If $a^x = 1$ where $a > 0$, $a \neq 1$, then $x = 0$
- If $a^x = b^x$ where $a > 0$, $b > 0$, $a \neq b$, then $x = 0$

Example 4

Solve each of the following equations in $\mathbb{R}$:

1 $8^{3x^2-12} = 1$ 2 $5^{x-4} = 2^x \times 2^{-x}$ 3 $2^{x-7} = 3^{7-x}$

Solution

1 $\because 8^{3x^2-12} = 1$ $\therefore 3x^2-12 = 0$ $\therefore 3x^2 = 12$

$\therefore x^2 = 12 \div 3$ $\therefore x^2 = 4$ $\therefore x = \pm\sqrt{4} = \pm 2$

---

2 $\because 5^{x-4} = 2^x \times 2^{-x}$ $\therefore 5^{x-4} = 2^{x-x}$ $\therefore 5^{x-4} = 2^0$

$\therefore 5^{x-4} = 1$ $\therefore x-4 = 0$ $\therefore x = 4$

---

3 $\because 2^{x-7} = 3^{7-x}$ $\therefore 2^{x-7} = \left(\left(\frac{1}{3}\right)^{-1}\right)^{7-x}$

$\therefore 2^{x-7} = \left(\frac{1}{3}\right)^{x-7}$ $\therefore x-7 = 0$ $\therefore x = 7$

TRY it Yourself 4

Solve each of the following equations in $\mathbb{R}$:

1 $9^{2x^2-2} = 4^2 \times 2^{-4}$ 2 $5^{8x-12} = \left(\frac{1}{5}\right)^{8x-12}$

Example 5

Find the value of x in each of the following :

1 $5^{x+1} + 5^{x+1} + 5^{x+1} = 75$

2 $8 \times 5^{3x-6} = 2^{3x-6} \times 125$

Solution

1 $\because 5^{x+1} + 5^{x+1} + 5^{x+1} = 75$

$\therefore 3 \times 5^{x+1} = 75$ $\quad \therefore 5^{x+1} = \frac{75}{3}$ $\quad \therefore 5^{x+1} = 25$

$\therefore 5^{x+1} = 5^2$ $\quad \therefore x + 1 = 2$ $\quad \therefore$ $x = 1$

---

2 $\because 8 \times 5^{3x-6} = 2^{3x-6} \times 125$

$\therefore 2^3 \times 5^{3x-6} = 2^{3x-6} \times 5^3$ $\quad \therefore \frac{5^{3x-6}}{5^3} = \frac{2^{3x-6}}{2^3}$

$\therefore 5^{3x-6-3} = 2^{3x-6-3}$ $\quad \therefore 5^{3x-9} = 2^{3x-9}$

$\therefore 3x - 9 = 0$ $\quad \therefore 3x = 9$ $\quad \therefore$ $x = 3$

TRY it Yourself 5

Find the value of x in each of the following :

1 $3^{2x-1} + 3^{2x-1} = 54$

2 $9 \times 2^{3x-7} = 4 \times 3^{3x-7}$

Example 6

Find the value of n in each of the following :

1 $\frac{6^{2n-3}}{2^{n-1} \times 3^{n-1}} = 6$

2 $\frac{(14)^{2n} \times 4^{n+1}}{4 \times 7^{2n-1} \times 16^n} = 7^n$

Solution

1 $\because \frac{6^{2n-3}}{2^{n-1} \times 3^{n-1}} = 6$ $\quad \therefore \frac{6^{2n-3}}{(2 \times 3)^{n-1}} = 6$

$\therefore \frac{6^{2n-3}}{6^{n-1}} = 6$ $\quad \therefore 6^{2n-3-n+1} = 6$

$\therefore 6^{n-2} = 6$ $\quad \therefore n - 2 = 1$ $\quad \therefore$ $n = 3$

---

2 $\because \frac{(14)^{2n} \times 4^{n+1}}{4 \times 7^{2n-1} \times 16^n} = 7^n$ $\quad \therefore \frac{(2 \times 7)^{2n} \times (2^2)^{n+1}}{2^2 \times 7^{2n-1} \times (2^4)^n} = 7^n$

$\therefore \frac{2^{2n} \times 7^{2n} \times 2^{2n+2}}{2^2 \times 7^{2n-1} \times 2^{4n}} = 7^n$ $\quad \therefore 2^{2n+2n+2-2-4n} \times 7^{2n-2n+1} = 7^n$

$\therefore 2^0 \times 7 = 7^n$ $\quad \therefore 7 = 7^n$ $\quad \therefore$ $n = 1$

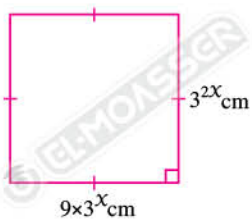
TRY it Yourself 6

Find the value of n if : $\frac{3^n \times 8^n}{(12)^{n+1}} = \frac{1}{3}$

Example 7

In the given figure :

A square. Find the length of its diagonal.

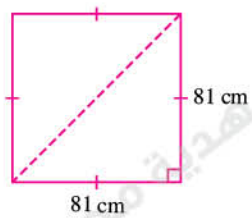


Solution

$\because$ The figure is a square.

$\therefore 9 \times 3^{x} = 3^{2x}$ $\quad\therefore 3^{2} \times 3^{x} = 3^{2x}$

$\therefore 3^{x+2} = 3^{2x}$ $\quad\therefore x + 2 = 2x$ $\quad\therefore$ $x = 2$



$\therefore$ The side length of the square $= 3^{2 \times 2} = 3^{4} = 81$ cm.

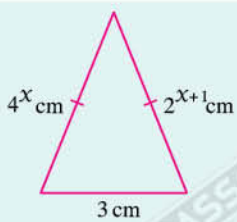
$\therefore$ The diagonal length $= \sqrt{81^{2} + 81^{2}} = 81\sqrt{2}$ cm.

TRY it Yourself 7

In the given figure :

An isosceles triangle.

Find its perimeter.



Free Part

Assessments and Final Revision

- Cumulative tests for each lesson.
- Important questions on each unit .
- School book final assessment.
- Final exams.



Exercise 2 Exponential Equations



Remember · Understand · Apply · Problems Solving | From the school book

Interactive test

1 Solve each of the following equations in $\mathbb{R}$:

1 $3^{2x+1} = 3^9$

2 $3^{x+1} = 81$

3 $5^{2x-1} = 125$

4 $4^{1-2x} = 2$

5 $25^{x+1} = 125$

6 $8^{x-1} \times 4^{2x+1} = 64$

2 Solve each of the following equations in $\mathbb{R}$:

1 $2^{x-7} = \frac{1}{32}$

2 $3^{2x-3} = \frac{1}{27}$

3 $\left(\frac{1}{3}\right)^{2x-1} = 243$

4 $\left(\frac{3}{5}\right)^{2x+1} = \frac{125}{27}$

5 $\left(\frac{1}{8}\right)^{x-2} = 128$

6 $2^{-3m+2} \times 4^{2m} = \frac{1}{8}$

3 Solve each of the following equations in $\mathbb{R}$:

1 $7^{2x+3} = 7^{x-4}$

2 $9^{3x+2} = 3^{x-1}$

3 $32^{x-3} = 8^{2x+1}$

4 $(27)^{2x-4} = 9^{2x}$

5 $64^{x+1} = 4^{x+7}$

6 $\left(\frac{1}{4}\right)^{2x-1} = 8^{x+3}$

7 $\left(\frac{3}{4}\right)^{5x+1} = \left(\frac{9}{16}\right)^{3x-2}$

8 $\left(\frac{3}{2}\right)^{5-3x} = \left(\frac{4}{9}\right)^{2x-7}$

4 Find the value of x in each of the following :

1 $16^x = 1$

2 $7^{x+2} = 2^x \times 2^{-x}$

3 $4^{x^2-9} = 1$

4 $2^{x^2} \times 2^x = 1$

5 $(5^{x-3})^{x+4} = 1$

6 $9^{x^2-2x+1} = \frac{25^x}{5^{2x}}$

5 Find the value of x in each of the following :

1 $7^x = 5^x$

2 $2^{x+1} = 3^{x+1}$

3 $\left(\frac{1}{2}\right)^{x^2-4} = 2^{x^2-4}$

4 $\left(\frac{1}{2}\right)^{2x-1} = 5^{1-2x}$

5 $6^{3x-12} = 3^{3x-12} \times 3^{3x-12}$

6 $9 \times 5^{x-1} = 25 \times 3^{x-1}$

6 Find the solution set in $\mathbb{R}$ for each of the following :

1 $2^{|x|} = 32$

2 $3^x + 3^x + 3^x + 3^x = 36$

3 $5^{x^2} = 625$

4 $2^{x^2} = 16^{3-x}$

5 $\frac{2^{x^2}}{16^x} = 32$

6 $(3^{x-2})^{x+2} = 27^x$

7 **Find the value of n in each of the following :**

1) $\frac{8^{3n} \times 16^{n+1}}{32^{2n}} = 2^7$

2) 📖 $\frac{36^{n+1}}{6^{n-1}} = \frac{1}{216}$

3) 📖 $\frac{8^n \times 9^n}{18^n} = 64$

4) $\frac{2^{2n} \times 9^{2-n}}{36^n} = 1$

5) 📖 $\frac{12^{n-1}}{2^{n-1} \times 3^{n-1}} = 1$

6) 📖 $\frac{2^n \times 9^{n+1}}{18^n} = 3^n$

8 📖 If $\frac{49^n \times 25^{2n} \times 3^{4n}}{7^{-n} \times 15^{4n}} = 343$, **then find the value of :** 6^{2n} « 36 »

9 If $\frac{6^{2n} \times 2^{2n}}{4^{2n} \times 3^{2n+4}} = 9^{-x}$, **then find the value of :** x « 2 »

10 If $\frac{7^x \times 6^x}{14^x} = 3^{2-y}$, **then find the value of :** $x + y$ « 2 »

11 If $3^x = 27$, $4^{x+y} = 1$, **find :** x , y « 3 , – 3 »

12 📖 **Volumes : In the given figure :**

A glass box is in the shape of a cube. The lengths of the sides of its base are 2^{x+1} cm , 4^x cm as shown in the figure

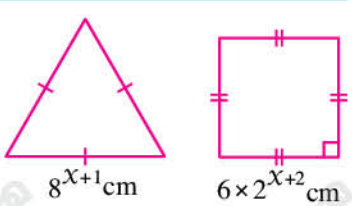
Find its volume in cubic centimeters.



« 64 cm^3 »

13 📖 **In the given figure :**

If the perimeter of the equilateral triangle is equal to the perimeter of the square , what is the side length of each shape ?



« 64 cm , 48 cm »

14 A square with a side length of 4^x cm , a rectangle with dimensions 4 cm , 2^{2x} cm.

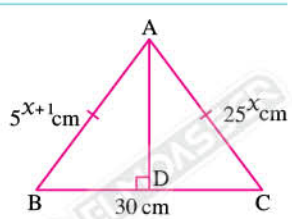
If the area of the square equals the area of the rectangle , find the area of each of them.

« 16 cm^2 , 16 cm^2 »

15 **In the given figure :**

ABC is an isosceles triangle , $\overline{AD} \perp \overline{BC}$

Find its area.



« 300 cm^2 »

16 📖 **In the given figure :**

A cardboard box is in the shape of a cuboid.

Its volume is 1 m^3.

Find the dimensions of the box.



« 1 m , 2 m , $\frac{1}{2}$ m »

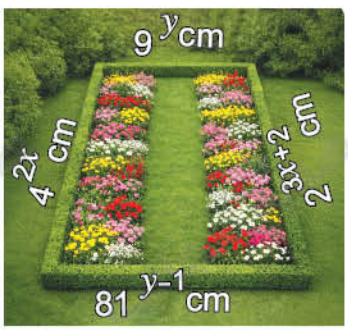
17 📖 **Gardens :**

A flower bed is in the shape of a rectangle.

Its dimensions are shown in the figure.

Calculate the perimeter of the flower bed.

« 674 cm »



18 📖 **Decoration : In the given figure :**

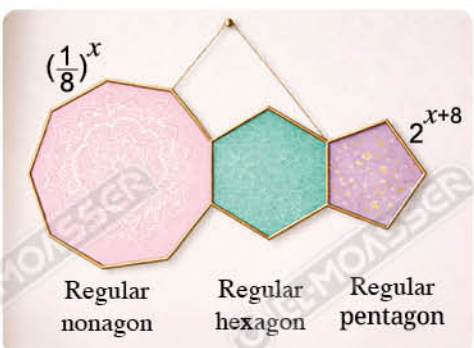
A decorative shape is hanged on a wall.

It consists of three regular polygons ,

a pentagon , a hexagon and a nonagon.

What is the value of x ?

« – 2 »



19 📖 **Discuss :** Marwan and Mariam both solved the equation :

$16^{x+1} = 64^{x-1}$

Which one followed the correct method of solution ?

Marwan's solution :

$16^{x+1} = 64^{x-1}$

$(2^4)^{x+1} = (2^6)^{x-1}$

$2^{4x+4} = 2^{6x-6}$

$4x + 4 = 6x - 6$

$4 + 6 = 6x - 4x$

$10 = 2x \quad x = 5$

Mariam's solution :

$16^{x+1} = 64^{x-1}$

$(4^2)^{x+1} = (4^3)^{x-1}$

$4^{2x+2} = 4^{3x-3}$

$2x + 2 = 3x - 3$

$3x - 2x = 2 + 3$

$x = 5$

Creative Thinking

20 Find the value of x in each of the following :

1 📖 $2^{x} + 2^{x+1} + 2^{x+2} = 112$ « 4 »

2 📖 $2 \times 3^{x+1} - 3^{x} = 5^{x+1} - 2 \times 5^{x}$ « 1 »

3 $\left(\left((2^{x})^{2}\right)^{2}\right)^{2} = 2^{x^2+12}$ « 2 or 6 »

Multiple Choice Questions



Choose the correct answer from the given ones :

1 If $2^{x-3} = 1$, what is the value of x ?
(a) 0 (b) 1
(c) 2 (d) 3

2 If $2^{x} = 8$, what is the value of x ?
(a) 1 (b) 2
(c) 3 (d) 4

3 If $3^{-x} = 3$, what is the value of x ?
(a) -3 (b) -1
(c) 1 (d) 3

4 If $3^{x+2} = 3^{2x-1}$, what is the value of x ?
(a) 2 (b) 3
(c) 4 (d) 5

5 If $7^{x-1} = 7^{1-x}$, what is the value of x ?
(a) -1 (b) 0
(c) 3^0 (d) 7

6 The solution set of the equation :
$(3^{x})^{x} = 81$ in $\mathbb{R}$ is .....
(a) $\{9\}$ (b) $\{9, -9\}$
(c) $\{2\}$ (d) $\{2, -2\}$

7 If $5^{x-1} = 3^{x-1}$, what is the value of x ?
(a) 0 (b) 1
(c) 3 (d) 5

8 If $9^{x} = 27$, what is the value of x ?
(a) $\frac{2}{3}$ (b) $\frac{3}{2}$
(c) $\frac{1}{3}$ (d) 3

9 What is the value of x that satisfies :
$(3^{x})^{2} + 5^{x+1} = 9^{x} + 25^{x-1}$?
(a) -2 (b) 1
(c) 5 (d) 3

10 If $5^{2x-1} = \left(\frac{1}{5}\right)^{-7}$, what is the value of x ?
(a) 2 (b) 3
(c) 4 (d) 5

11 If $3^{x} + 3^{x} = 6$, what is the value of x ?
(a) 1 (b) 2
(c) 3 (d) 9

12 If $8 \times 5^{x+1} = 125 \times 2^{x+1}$, what is the value of x ?
(a) 0 (b) 1
(c) 2 (d) 3

13 If $\left(\frac{3}{4}\right)^{3x} = \left(\frac{64}{27}\right)^{-1}$, what is the value of x ?
(a) -1 (b) 3
(c) -3 (d) 1

14 What is the solution set of the equation :
$\left(\frac{1}{2}\right)^{3-x} = 4^{2x+3}$ in $\mathbb{R}$?
(a) $\{3\}$ (b) $\{3, -3\}$
(c) $\{-3\}$ (d) $\{2, 3\}$

Unit 1 Summary

Laws of Exponents in Real Numbers

1 Multiplying two powers with the same base

$a^m \times a^n = a^{m+n}$

When multiplying two powers with the same base, add the exponents.

For example :

$2^3 \times 2^4 = 2^{3+4} = 2^7$

2 Dividing two powers with the same base

$a^m \div a^n = a^{m-n}$

When dividing two powers with the same base, subtract the exponents.

For example :

$2^5 \div 2^2 = 2^{5-2} = 2^3$

3 Power of a product of two numbers

$(a \times b)^n = a^n \times b^n$

The exponent of the product is distributed to each factor.

For example :

$(2 \times 3)^4 = 2^4 \times 3^4$

4 Power of a quotient of two numbers

$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$

The exponent of the quotient is distributed to the numerator and the denominator.

For example :

$\left(\frac{2}{3}\right)^3 = \frac{2^3}{3^3}$

5 Power of a power rule

$(a^m)^n = a^{m \times n}$

To find the power of a power, multiply the exponents.

For example :

- $(2^3)^2 = 2^{3 \times 2} = 2^6$
- $(5^2)^4 = 5^{2 \times 4} = 5^8$

6 Power of a product of two powers rule

$(a^m \times b^n)^k = a^{mk} \times b^{nk}$

For example :

$(5^2 \times 2^3)^2$

$= 5^{2 \times 2} \times 2^{3 \times 2}$

$= 5^4 \times 2^6$

7 Power of a quotient of two powers rule

$\left(\frac{a^m}{b^n}\right)^k = \frac{a^{mk}}{b^{nk}}$

For example :

$\left(\frac{2^5}{3^2}\right)^3 = \frac{2^{5 \times 3}}{3^{2 \times 3}}$

$= \frac{2^{15}}{3^6}$

8 The zero exponent

$a^0 = 1$

Any non-zero real number raised to the power of zero equals 1

For example :

- $5^0 = 1$
- $(-3)^0 = 1$

9 The negative exponent

$a^{-n} = \frac{1}{a^n}$

The negative exponent means the reciprocal of the number raised to the positive exponent.

For example :

- $2^{-3} = \frac{1}{2^3}$
- $(-4)^{-2} = \frac{1}{(-4)^2}$

10 The negative exponent of a fraction

$\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n$

The negative exponent of a fraction means the reciprocal of the fraction raised to the positive exponent.

For example :

$\left(\frac{2}{3}\right)^{-2} = \left(\frac{3}{2}\right)^2$

The Exponential Equation

1 Meaning of the exponential equation

An exponential equation is an equation that contains a variable in the exponent.

General form

$a^x = b$

where $a > 0$, $a \neq 1$

Examples of exponential equations :

- $2^x = 8$
- $5^{2x} = 125$
- $3^{x+1} = 27$
- $4^x = 2^{x+2}$

2 Solving the exponential equation

To solve the exponential equation, use the property of equality of powers.

If $a^x = a^y$, then $x = y$
$a > 0$, $a \neq 1$

Meaning: If two powers with the same base are equal, then their exponents are equal.



Methods of solution

First Method : Same base

If the equation is in the form : $a^x = a^y$

$(a > 0 , a \neq 1)$

Same base, therefore equate the exponents $x = y$

Example 1

Solve the equation :

$2^x = 16$

Solution

$\because 2^x = 16$

$\therefore 2^x = 2^4$

$\therefore x = 4$

Example 2

Solve the equation :

$3^{x+2} = 81$

Solution

$\because 3^{x+2} = 81$

$\therefore 3^{x+2} = 3^4$

$\therefore x + 2 = 4$

$\therefore x = 2$

Second Method : Unifying the base

If the equation has two different bases, convert it to the same base, then equate the exponents.

Example 3

Solve the equation :

$4^x = 2^{x+2}$

Solution

$\because 4^x = 2^{x+2}$

$\therefore (2^2)^x = 2^{x+2}$

$\therefore 2^{2x} = 2^{x+2}$

$\therefore 2x = x + 2$

$\therefore x = 2$

Various additional examples

Example 4

Solve the equation :

$5^{2x-1} = 125$

Solution

$\because 5^{2x-1} = 125$

$\therefore 5^{2x-1} = 5^3$

$\therefore 2x - 1 = 3$

$\therefore 2x = 4$

$\therefore x = 2$

Example 5

Solve the equation :

$9^{x-1} = \frac{1}{27}$

Solution

$\because 9^{x-1} = \frac{1}{27}$

$\therefore (3^2)^{x-1} = 3^{-3}$

$\therefore 3^{2x-2} = 3^{-3}$

$\therefore 2x - 2 = -3$

$\therefore 2x = -1$

$\therefore x = -\frac{1}{2}$

Notes

- If $a^x = 1$ where $a > 0$, $a \neq 1$, then $x = 0$
- If $a^x = b^x$ where $a > 0$, $b > 0$, $a \neq b$, then $x = 0$

Example 6

1 If $7^{x-5} = 1$

$\therefore x - 5 = 0$ $\therefore x = 5$

2 If $2^{x+4} = 5^{x+4}$

$\therefore x + 4 = 0$ $\therefore x = -4$

Assessment of unit one from the school book

► Choose the correct answer from the given ones :

1 Which of the following is equal to $(a^2)^4 \times a^5$?

(a) a^{11} (b) a^{13} (c) a^{40} (d) a^{30}

2 If $\frac{3^x \times 27^x}{81^x} = 2^{-2x}$, what is the value of x ?

(a) 0 (b) 1 (c) 2 (d) – 1

3 If $3^x = b$, which of the following is equal to 9^{x+1} ?

(a) $27\ b^2$ (b) $3\ b^2$ (c) $9\ b^2$ (d) $3\ b$

4 If $4^x = 8$, what is the value of x ?

(a) 1 (b) 2 (c) $\frac{2}{3}$ (d) $\frac{3}{2}$

► Complete each of the following :

5 If $7^{x+1} = 5^{x+1}$, then $x =$ .....

6 The result of $\frac{(-2a^2)^3}{2a}$ in its simplest form is .....

7 If $5^{x-3} = 1$, then $2^x =$ .....

8 If $2^a + 2^a + 2^a = 24$, then $a =$ .....

► Answer each of the following questions :

9 If $2^x = 5$, **then find the value of :** $8^x - 3 \times 2^{x+2}$

10 **Simplify to its simplest form :** $\frac{4^x \times 9^{2-x}}{6^{2x}}$, then find the value of the result when $x = 1$

11 **Find in $\mathbb{R}$ the solution set of the equation :** $5^{x^2-4} = 125^x$

12 **Find the value of m if :** $3^{-2m+1} \times 9^{2m} = \frac{1}{27}$

Model Exam On unit ONE

First Group

▶ **Choose the correct answer from the given ones :**

1 If $3^{x} = 5$, what is the value of 9^{x} ?

(a) 15 (b) 25 (c) 45 (d) 81

---

2 If $4^{2x} = \frac{1}{8}$, what is the value of x ?

(a) $\frac{1}{4}$ (b) $-\frac{1}{4}$ (c) $-\frac{3}{4}$ (d) $\frac{3}{4}$

---

3 $(x^{2})^{3} + (x^{4})^{0} =$ .....

(a) x^{46} (b) x^{10} (c) x^{6} (d) $x^{6} + 1$

---

4 Which of the following equals $\frac{(b^{2})^{3} \times (b^{3})^{4}}{(b^{5})^{2}}$?

(a) b^{5} (b) b^{6} (c) b^{8} (d) b^{28}

---

5 If $7^{x-2} = 1$, what is the value of 5^{x-3} ?

(a) 5 (b) $\frac{1}{5}$ (c) 25 (d) 1

---

6 If $5^{x} + 5^{x} = 250$, what is the value of x ?

(a) 3 (b) 2 (c) 1 (d) 0

Second Group

▶ **Answer the following questions :**

1 **Prove that :** $\frac{(25)^{x+1} \times 9^{x}}{(15)^{2x-1}} = 375$

---

2 If $3^{x} = 4$, **then find the value of :** $9^{x} + 3^{x+1} + 3^{-x}$

---

3 **Find in $\mathbb{R}$ the solution set of the equation :** $2^{x^{2}} = 16^{3-x}$

4 If $3^{2x-1} = 27$, $\left(\sqrt{7}\right)^{x+y} = 49$, **find :** $\left(\sqrt{5}\right)^{x-y}$

---

5 A rectangle with dimensions 2^{x+1} cm , 11 cm and its area equals 176 cm^2 , what is the value of x ?

---

6 **Find in $\mathbb{R}$ the solution set of the equation :** $8 \times 3^{x-2} = 27 \times 2^{x-2}$

---

7 **Simplify :** $\dfrac{(32)^{x} \times 3^{2x+1}}{6^{2x+1} \times \left(\sqrt{2}\right)^{-4x}}$, then find the value of the result when $x = 1$

---

8 **Solve the following equation in $\mathbb{R}$:** $7^{x+2} = 2^{x} \times 2^{-x}$